

NAME: SANTHOSH D

REG NO: (192412209)

COURSE NAME: ANALOG AND DIGITAL COMMUNICATION (ECA0802)

EX NO: 10

CALCULATION OF PROCESSING GAIN AND GENERATION OF
PN SEQUENCES FOR SPREAD SPECTRUM COMMUNICATION
USING MATLAB SIMULATION

Objectives

- i) To calculate the processing gain for various PN sequence lengths and tabulate the corresponding processing gain values.
- ii) To visualize and compare the processing gain obtained for different PN sequence lengths using MATLAB plots and analyze the effect of sequence length on spread spectrum performance.

PROGRAM (MATLAB Code)

```
clc;

clear;

close all;

% PN sequence lengths
PN_length = [7 15 31 63 127 255 511 1023];

% Processing Gain (Linear)
PG_linear = PN_length;

% Processing Gain in dB
PG_dB = 10*log10(PG_linear);

% Display table
T = table(PN_length',PG_dB',...
'VariableNames',{'PN_Length','Processing_Gain_dB'});

disp('Processing Gain Table');

disp(T);
```

```
% Plot
```

```
figure;
```

```
stem(PN_length,PG_dB,'filled');
```

```
grid on;
```

```
xlabel('PN Sequence Length');
```

```
ylabel('Processing Gain (dB)');
```

```
title('Spread Spectrum: Processing Gain vs PN Sequence Length');
```

OUTPUT:

